

Business Problem Statement

A prominent retail organization aims to gain a deeper understanding of its customers' purchasing behavior to enhance sales performance, customer satisfaction, and long-term loyalty. The management team has observed shifts in buying patterns across different customer demographics, product categories, and sales channels (online and offline). They are especially interested in identifying how factors such as discounts, customer reviews, seasonal trends, and payment preferences influence purchasing decisions and repeat buying behavior.

You are required to analyze the company's consumer behavior dataset to address the following overarching business question:

"How can the company utilize consumer shopping data to recognize trends, strengthen customer engagement, and optimize its marketing and product strategies?"

Deliverables

- 1. Data Preparation & Modeling (Python):** Clean, preprocess, and transform the raw dataset to make it suitable for analysis.
- 2. Data Analysis (SQL):** Structure the data appropriately, simulate business transactions, and execute SQL queries to uncover insights related to customer segments, loyalty, and the key drivers of purchases.
- 3. Visualization & Insights (Power BI):** Develop an interactive dashboard that showcases important trends and patterns, allowing stakeholders to make informed, data-driven decisions.
- 4. Report:** Prepare a concise project report summarizing the major findings and business recommendations.
- 5. GitHub Repository:** Include all Python scripts, SQL queries, and Power BI dashboard files in a well-organized repository.